Jilei Lin

Ph.D. in Statistics, The George Washington University
Arlington, VA, U.S.

jileil2@gwmail.gwu.edu
ORCID: 0000-0002-7147-9851
Google Scholar | GitHub

Research Profile

I develop statistical methodology for heterogeneous, spatially dependent, and censored data, with applications in medicine, environmental health, spatial transcriptomics, and social science.

Core methods: Quantile regression; distributed inference; spatial analysis; dimension reduction; survival analysis; missing data analysis; longitudinal data analysis; transfer learning.

Education

2021–2026	**Ph.D. in Statistics**, The George Washington University, Washington, D.C. Dissertation: *Quantile Regression for Spatial Data and Censored Data*. Advisors: Huixia Judy Wang and Feifang Hu.
2019–2021	**M.S. in Statistics**, University of Illinois Urbana-Champaign, IL.
2015–2019	**B.S. in Finance**, *summa cum laude*, Kean University, Union, NJ.

Publications

Published and Accepted

[1] **Smoothed Quantile Regression for Spatial Data**.
Lin, J., Wang, H., Wang, L., and Kim, M. **Accepted**; *Journal of Computational and Graphical Statistics (JCGS)*. Software: jileil2/SQBiT.

[2] **Hybrid Supervised-unsupervised Modeling for Post-hurricane Private Well Contamination Risk Score Using Empirical Validation and Community Groundtruthing**.
Lin, J.†, Zhang, J., Wei, E., Shea, K., Wang, H. J., Liddie, J. M., Hernandez, E., Norford, C., and Hu, X. C. **Accepted**; *GeoHealth*. doi:10.1029/2026GH001858.

[3] **Minimizing Post-shock Forecasting Error through Aggregation of Outside Information**.
Lin, J. and Eck, D. J. (2021). *International Journal of Forecasting*, 37(4), 1710–1727. doi:10.1016/j.ijforecast.2021.03.010.

[4] **The Application of PROMETHEE Multi-criteria Decision Aid in Financial Decision Making: Case of Distress Prediction Models Evaluation**.
Mousavi, M. M. and **Lin, J.** (2020). *Expert Systems with Applications*. doi:10.1016/j.eswa.2020.113438.

[5] **Identifying Opinion Leaders in Twitter during Social Events: Social Impact Theory from Influencer's Perspective**.
Lin, J., Wang, Z., Huang, Y., and Chen, R. (2020). *2019 4th International Conference on Communication and Information Systems*, 197–205. doi:10.1109/ICCIS49662.2019.00042.

†Corresponding author.

Under Review or Revision

[6] **Distributed Inference for Spatial Quantile Regression**.
Lin, J., Wang, H., and Wang, L. **Reject & Resubmit**; *Journal of the American Statistical Association: Theory and Methods (JASA T&M)*.

[7] **Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation**.
Liu, X., **Lin, J.**, Yan, H., Shi, H., Montellier, E., Chi, E. C., Hainaut, P., and Wang, W. **Major revision**; *Journal of the American Statistical Association: Applications and Case Studies (JASA A&CS)*.

[8] **Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis**.
Lin, J., Wang, H., and Luo, J. **Major revision**; *Annals of Applied Statistics (AoAS)*.

[9] **Striational Antibody-Associated Myositis–Bridging the Gap between Thymoma and Myasthenia Gravis: A Systematic Review**.

Luo, J., **Lin, J.**, Shymansky, J., and Wang, H. J. **Submitted**; *Nature Communications*. Code.

Working Papers

[10] **Shed Acetylcholine Receptors from Damaged Muscles Trigger Myasthenia Gravis in a Rodent Model**.
Lawson, B. S., Li, X., **Lin, J.**, Kuzma, L. C., Pham, A., Wu, Y., Borbi, A., Xu, Q., Wang, H., Garden, O. A., and Luo, J. To be submitted.

[11] **Quantile Regression for Spatial Survival Data**.
Lin, J. and Wang, H. J. Work in progress.

[12] **Predicting Drinking Water Contaminants in Underserved Areas Using Transfer Learning: An Informative Subset-Based Multitask Quantile Deep Neural Network Approach**.
Liddie, J. L., **Lin, J.**, Wei, E., Shea, K., Zhang, X., and Hu, X. Work in progress.

Software

SQBiT	R package implementing smoothed quantile regression for spatial data. Associated with the JCGS publication. jileil2/SQBiT.
DISQ	Distributed inference framework for spatial quantile regression. Associated with the JASA revision. jileil2/DISQ.
PFAS	Interactive Shiny application for screening post-hurricane private well contamination risk. PFAS Risk Screening Tool.

Professional Experience

2025–2026	**Research Assistant**, Predicting Drinking Water Contaminants in Underserved Areas with Transfer Learning. Principal investigators: Xindi Hu and Xiaoke Zhang.
2025	**Research Assistant**, Data-Driven Solutions to Mitigate the Impact of Hurricane on Private Well, Drinking Water Quality, and Human Health. Principal investigator: Xindi Hu.
2024–2025	**Research Assistant**, Unlocking Complex Heterogeneity in Large-Scale Spatial-Temporal Data with Adaptive Quantile Learning. Principal investigator: Huixia Judy Wang.
2023–2024	**Research Assistant**, Antigen-Specific Immunosuppression of Myasthenia Gravis by CAR-Engineered Tregs. Principal investigator: Jie Luo.

Teaching Experience

2024	Teaching Assistant, Introduction to Mathematical Statistics II (STAT 4158), Introduction to Business and Economic Statistics (STAT 1051), Mathematical Statistics II (STAT 6202), GWU.
2023	Teaching Assistant, Regression Analysis (STAT 2118), Survival Analysis (STAT 6227), Introduction to Business and Economic Statistics (STAT 1051), GWU.
2022	Teaching Assistant, Data Analysis (STAT 6210), Mathematical Statistics I (STAT 6201), GWU.
2021	Teaching Assistant, Advanced Regression Analysis (STAT 527), University of Illinois Urbana-Champaign.
2020	Teaching Assistant, Applied Regression and Design (STAT 425), University of Illinois Urbana-Champaign.

Presentations

2025	The Annual ASA/IMS Spring Research Conference on Statistics in Industry and Technology, New York, NY.
2025	The Past, Present and Future of Statistics in the Era of AI, Washington, D.C.
2024	ICSA Applied Statistics Symposium, Nashville, TN.
2024	Statistics in the Age of AI, Washington, D.C.
2023	ICSA Applied Statistics Symposium, Ann Arbor, MI.
2018	INFORMS International Conference, Taipei, Taiwan.

Mentoring

2024–2025 Maxwell Beveridge, undergraduate student.

Honors and Awards

2026	Minna Miran Kullback Memorial Prize for Leadership and Service, Department of Statistics, GWU.
2025	Ph.D. Student Poster Competition Award, StatConnect 2025, Vienna, VA.
2024	GW Statistics Cornfield Travel Award, Department of Statistics, GWU.

Skills

Programming	R, Python, Bash, LaTeX.
HPC and Automation	SLURM job scripting, GPU/CPU cluster workflows, environment modules.
Reproducibility	Git/GitHub project management, version control, workflow organization.

References

Huixia Judy Wang	Professor and Chair, Department of Statistics, Rice University. jw322@rice.edu.
Feifang Hu	Professor and Chair, Department of Statistics, The George Washington University. feifang@gwu.edu.
Lily Wang	Professor, Department of Statistics, George Mason University. lwang41@gmu.edu.